STUDY ON LIDAR SYSTEMS TO BUILD TOPOLOGIES FOR TSUNAMI DAMAGE CONTROL

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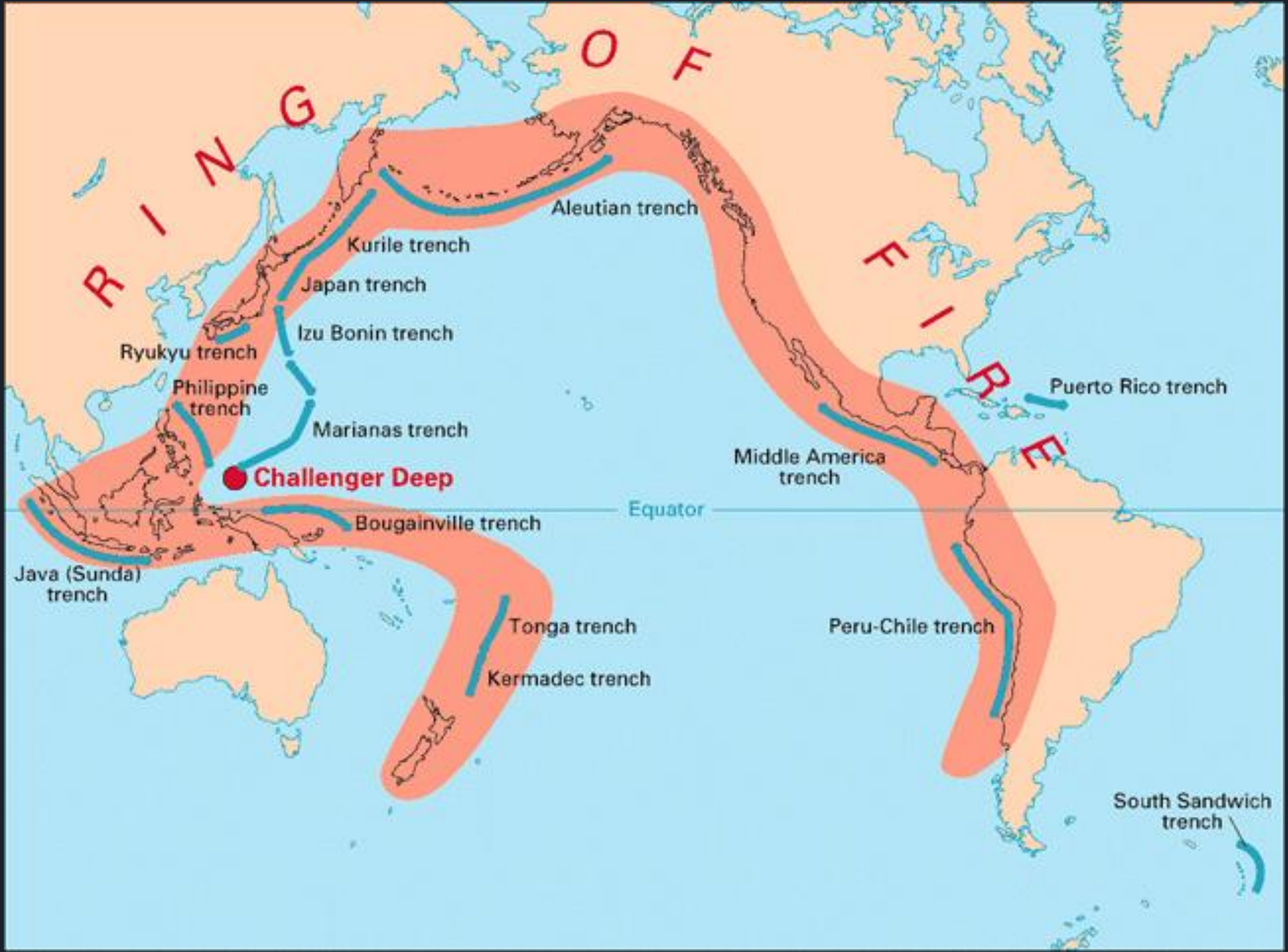
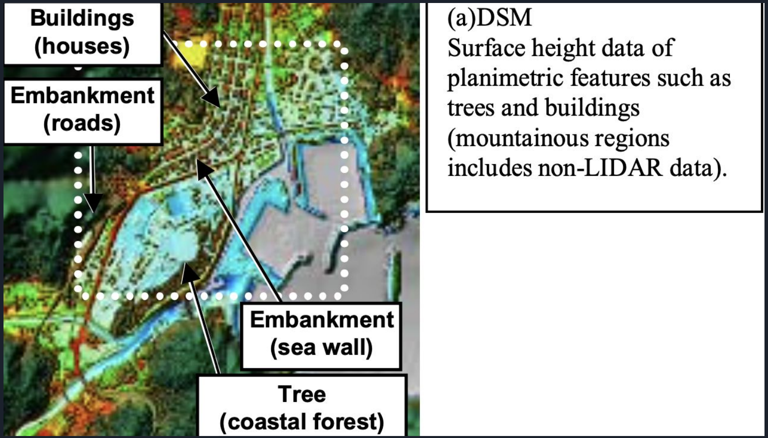
Why did we choose this project?

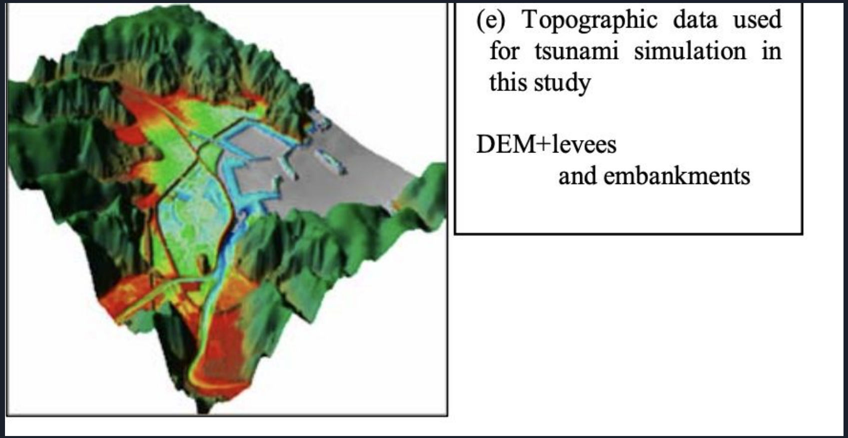
- Between the years 1996 to 2015 only 16 major tsunamis have been recorded,however these occurrences of tsunamis have claimed a death toll of 250,900 deaths
- The average death toll per event is 15,600 which is extremely high
- We need to build topologies that are capable of small measurement errors.(more specifically with an error margin of 30 cm horizontally and 15 cm vertically)


RING OF FIRE
Aleutian trench
Kurile trench
Japan trench
Izu Bonin trench
Ryukyu trench
Philippine trench
Marianas trench
Challenger Deep
Equator
Bougainville trench
Java (Sunda) trench
Tonga trench
Kermadec trench
Puerto Rico trench
Middle America trench
Peru-Chile trench
South Sandwich trench

WHAT WE ARE CAPABLE TO BUILD WITH OUR SYSTEM

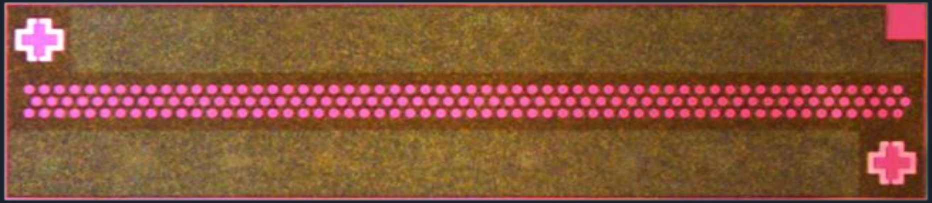
- Our objective is be able to build a LiDAR system capable of having an error of only ±30 cm horizontally and ±15cm



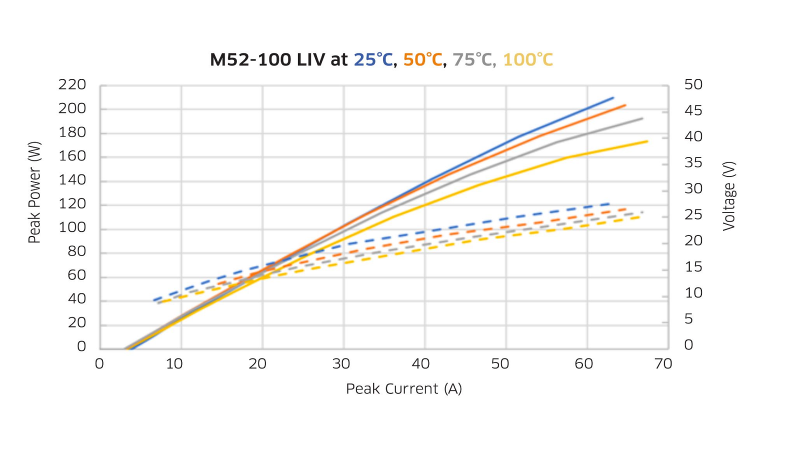


STATE OF THE ART TECHNOLOGIES UTILIZED

To increase the precision of our circuit we need high point density cloud. We need laser capable of generating fast pulses



Lumentum M52-100 Laser



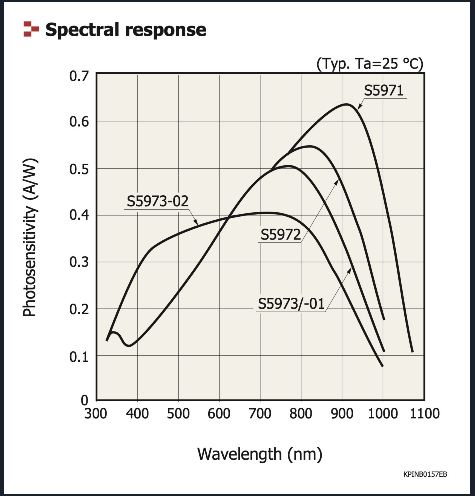
Spectral Response of M52-100

Given the need for a high-speed system, the design of the controller from scratch is complex. Therefore, we use EPC controllers to facilitate and speed up the development process

EPC9179

Employment of small rise and fall time photodiodes.

S5971 bY HAMAMATSU



Responsivity of S5917

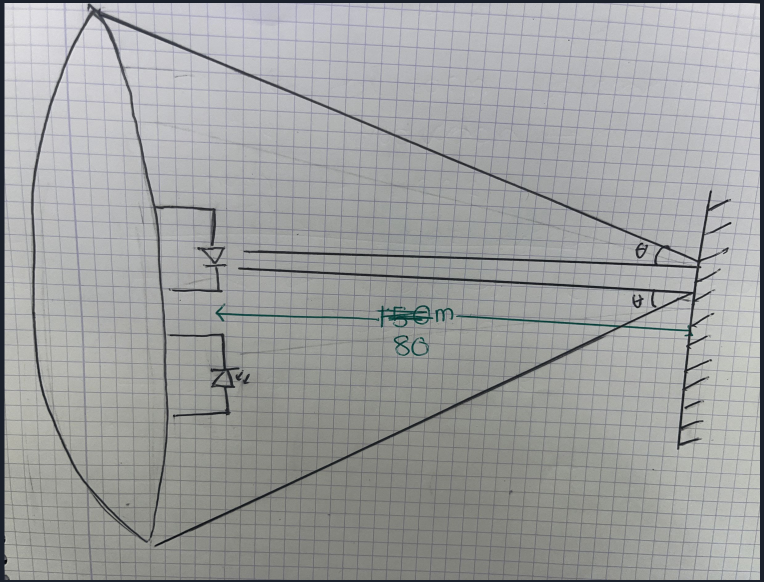
Reduction of complexity by deployment of drains with an integrated GPS system.

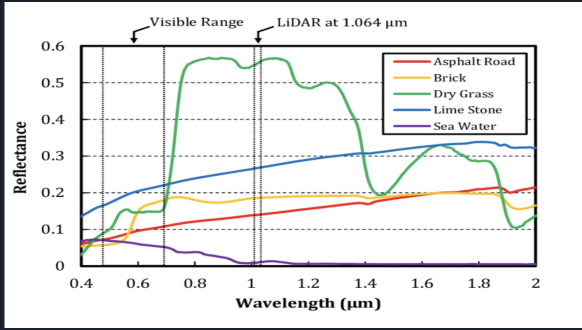
X500 from CHCNAV

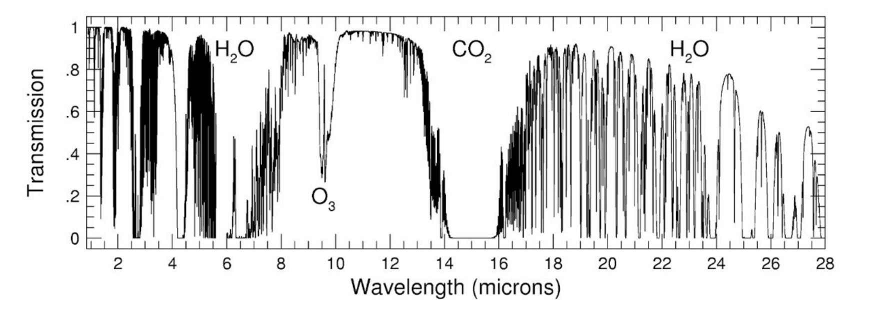
drones

OPTICAL POWER CONSIDERATION

The main cause of attenuation in our circuit is due to diffraction causing original pulses to be spread out and absorption by surfaces pulses are directed to







$$\frac{Prx}{Ptx} = C \cdot \frac{Arx}{\pi \cdot (\theta . d)^2}$$

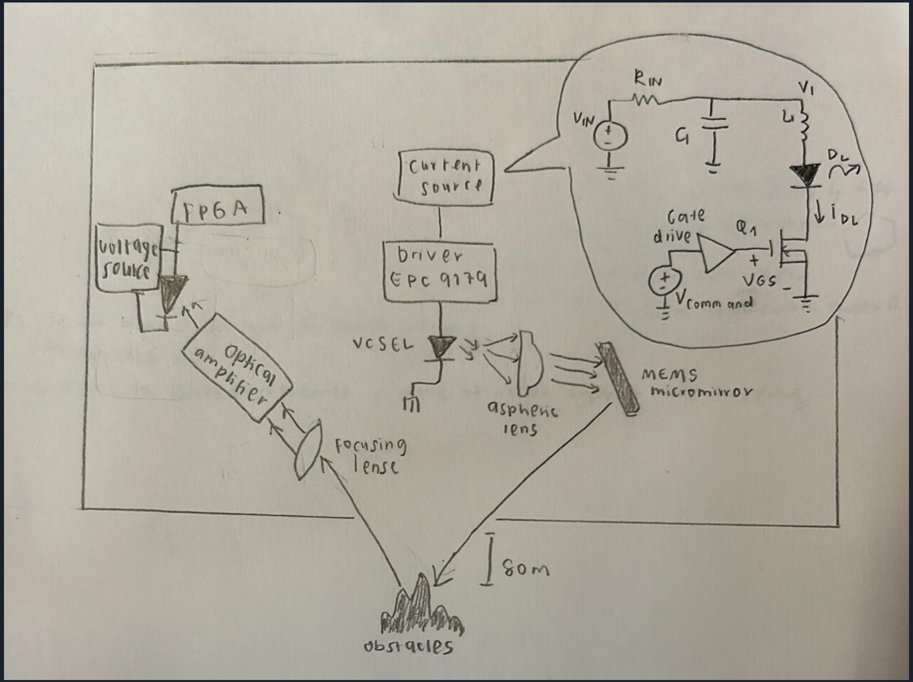
$$\frac{Prx}{Ptx} = 3.03 \times 10^{-15}$$


BOA930S

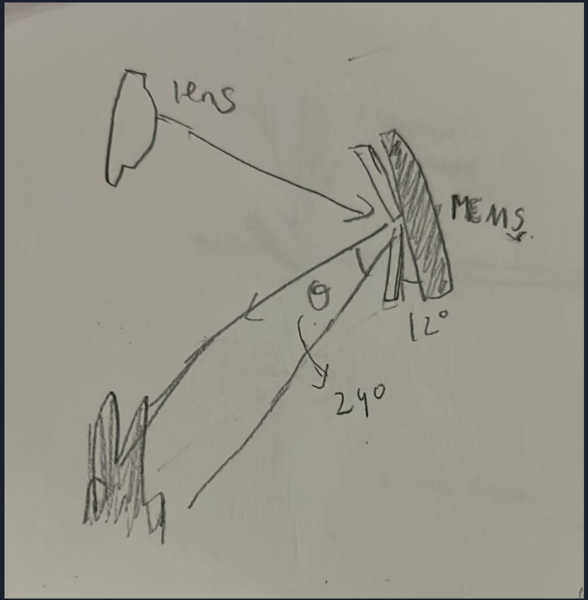
$$Pout = 10^{\frac{25}{10}} \times 7.27 \times 10^{-13} = 230\text{pW}$$

From this value of output power we can calculate the current generated by out photodiode.Its value is 150pA

DESIGN OF COMPLETE SYSTEM

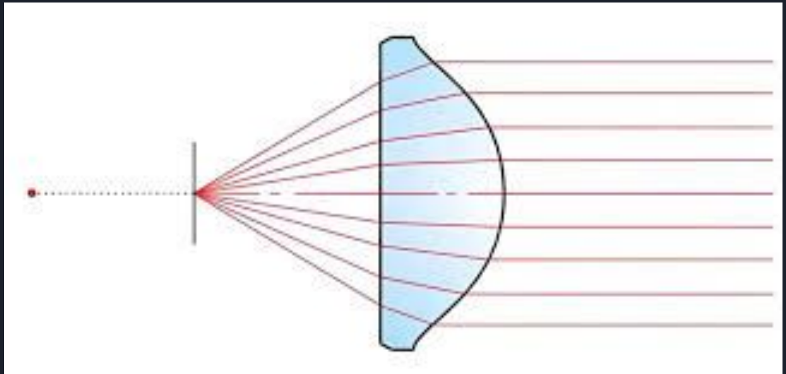


MEMS micromirrors as scanning device



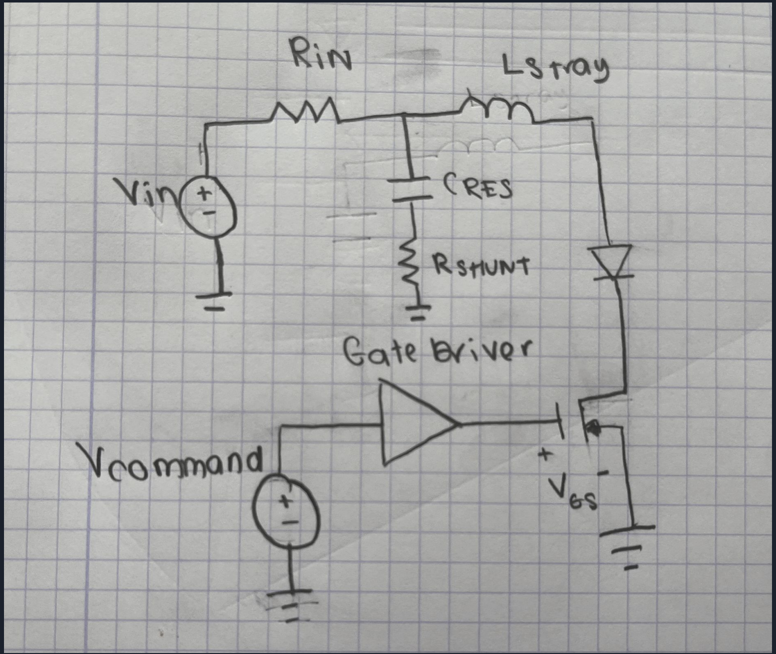
- a tiny, fast-moving mirrors that can acts as beam steering for precisely detecting obstacles within field of view (FOV).
- high reflectivity using aluminum foils.

ASPHERIC LENS



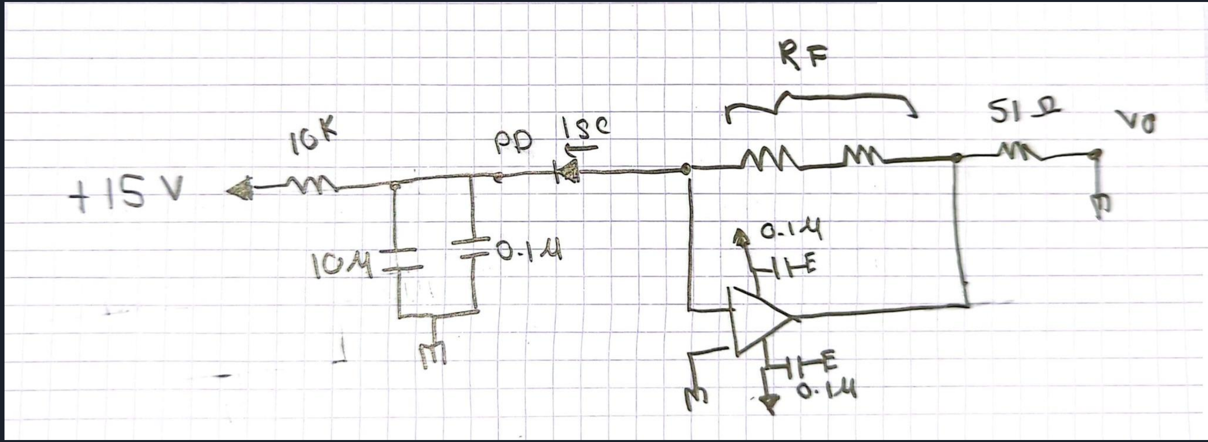
Placed about one focal length away from the sensor which turns light into parallel beam before it hits the micromirror

CURRENT DRIVER OF LASER



objective : to generate very fast and powerful pulses to drive the laser

RECEPTOR CIRCUIT



$$V0 = Rf \times Ipd = 2.3mV$$

ERROR SOURCES CONSIDERATION

$$Ix = \sqrt{2 \times e \times (Ifd + Idark) \times BW} = 1.3pA$$

$$Ith = \frac{4 \times Kb \times T \times BW}{Rf} = 8.7 \times 10^{-23} \simeq 0$$

$$SNR = \frac{Ipd}{Ix} = 115.3$$

CONCLUSION

- Tsunamis pose a severe threat due to their devastating impact, and because response time is limited, it's crucial to have precise terrain mapping.
- By integrating high-performance elements such as the M52-100 VCSEL laser array, EPC9179 pulse driver, and S5971 photodiode, our system achieves pulse widths down to nanoseconds, allowing us to reach resolution at the centimeter level.
- The outcome is a rapid and accurate mapping tool that supports tsunami simulation models and evacuation strategy planning, potentially saving lives.


Muchas gracias